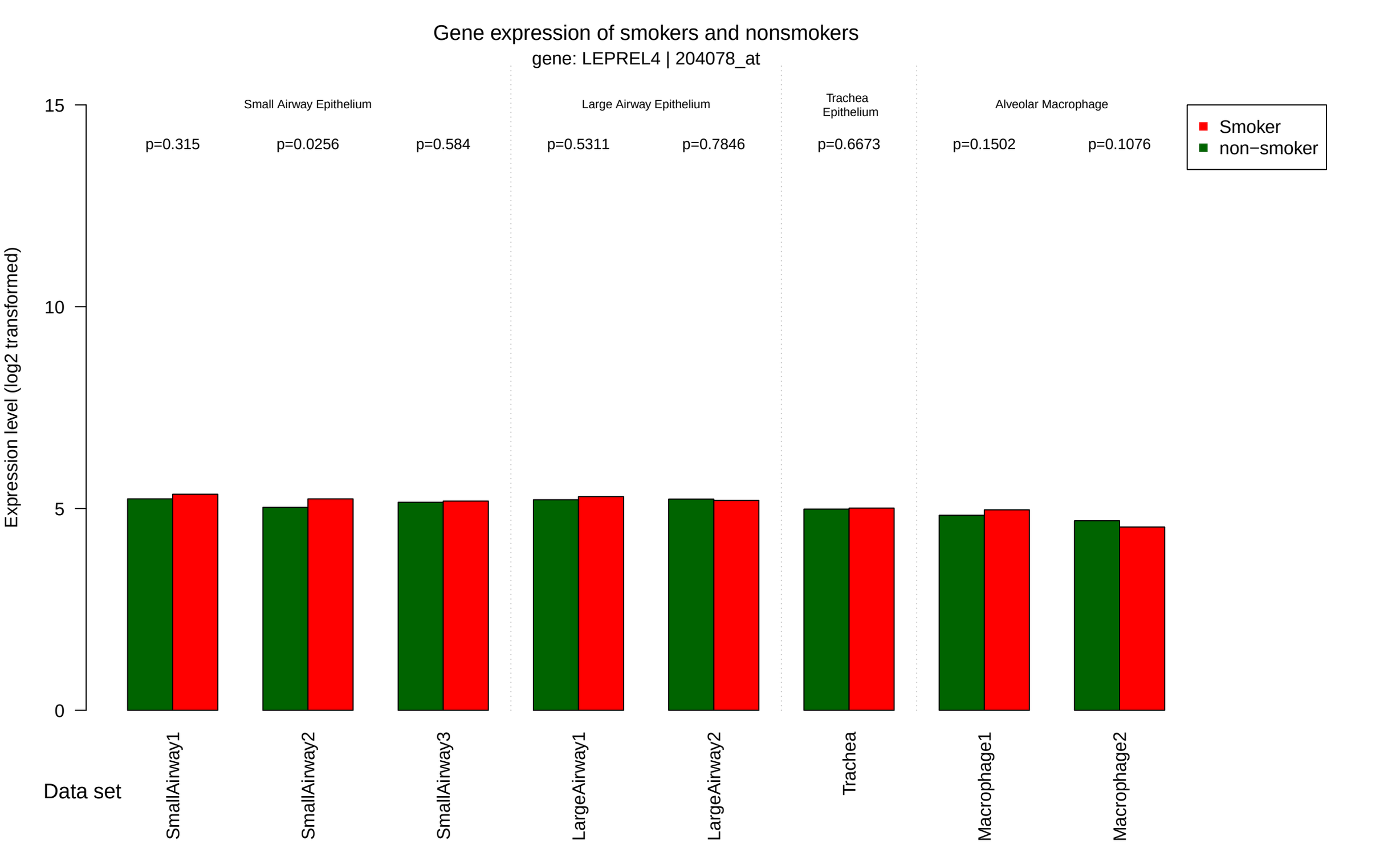
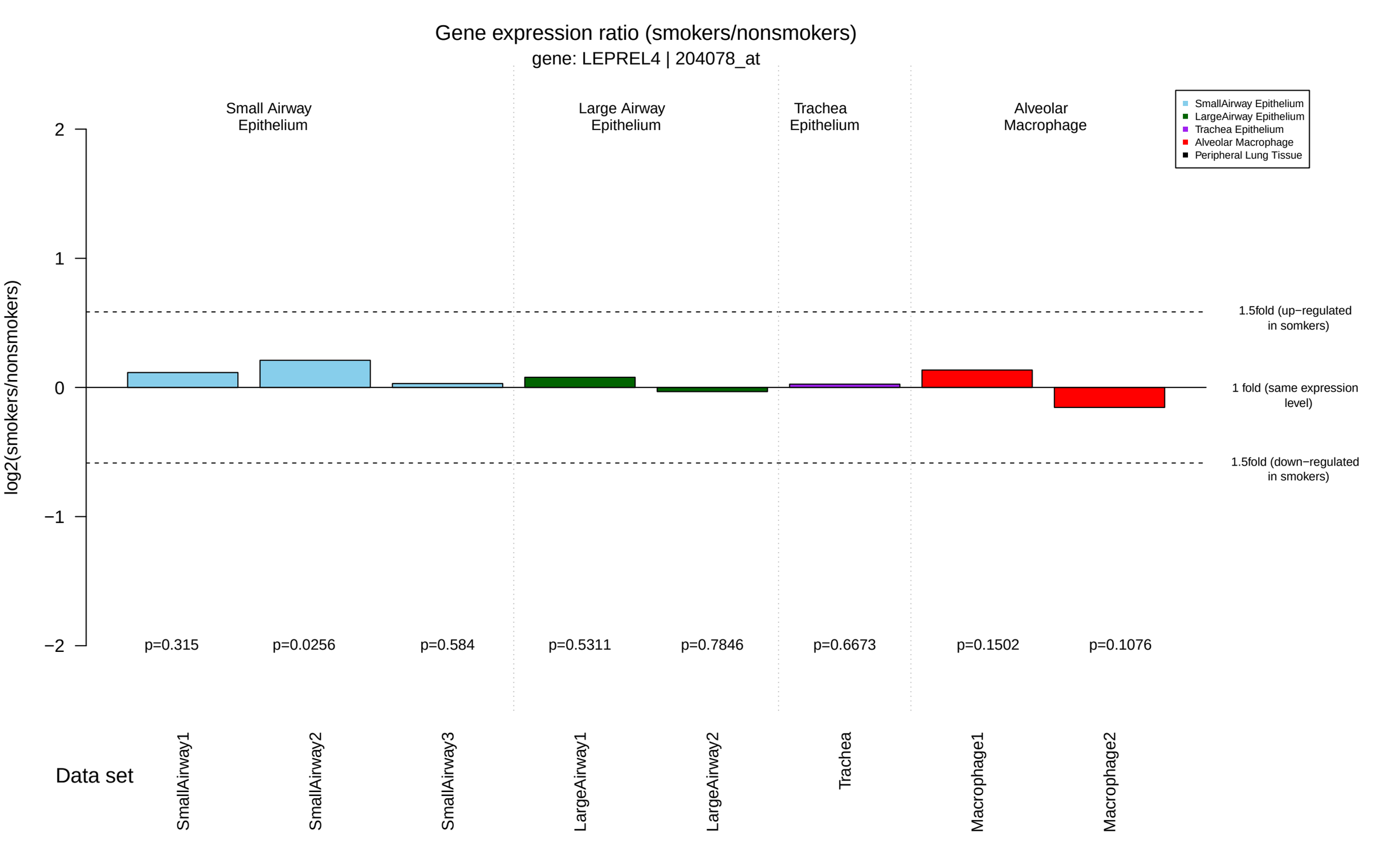
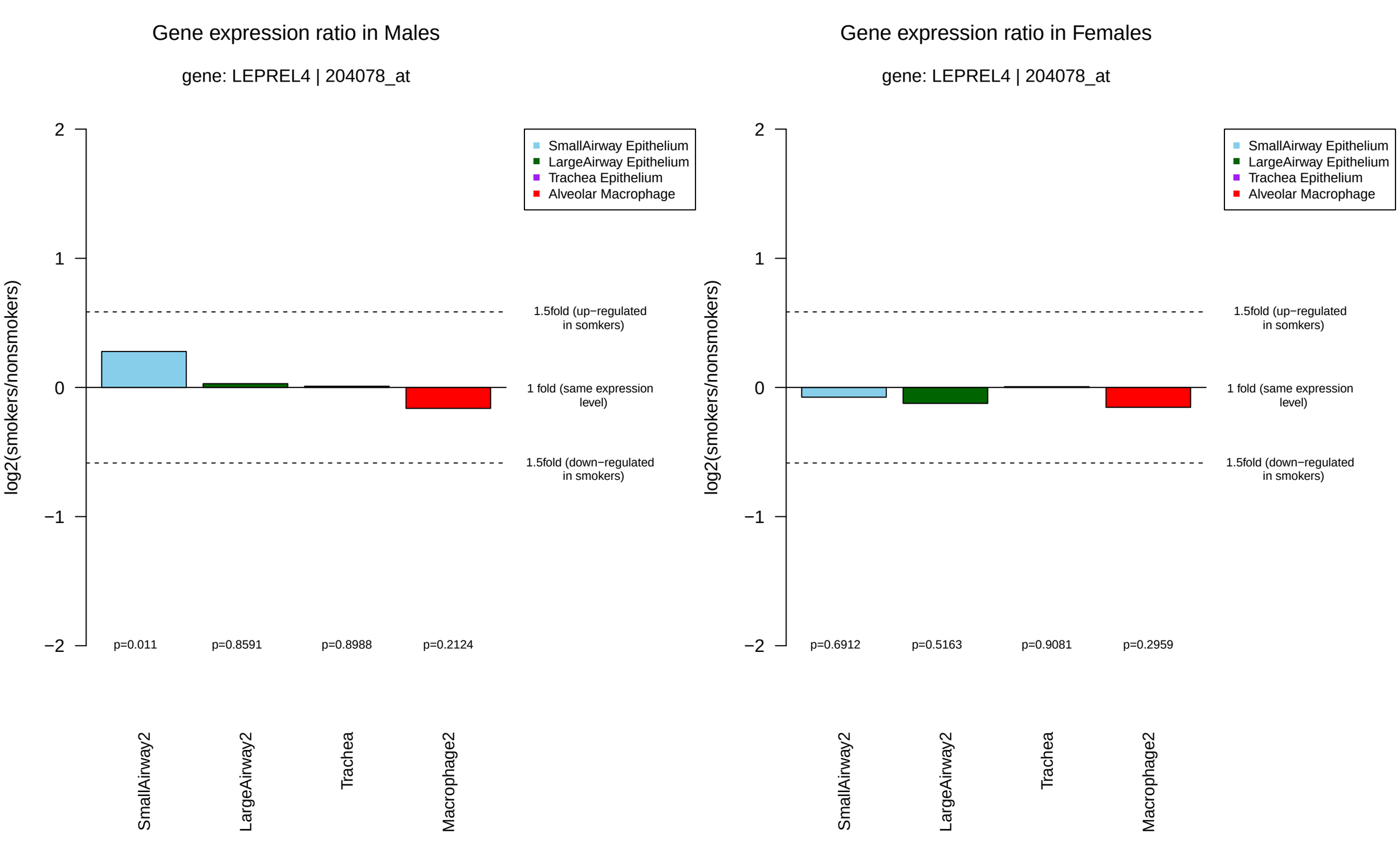
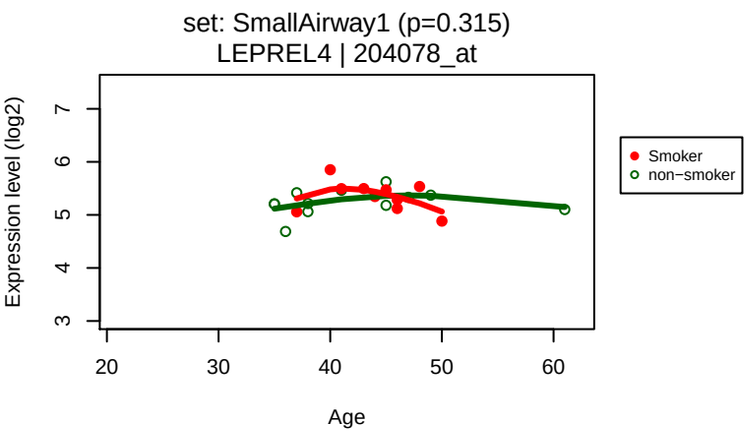
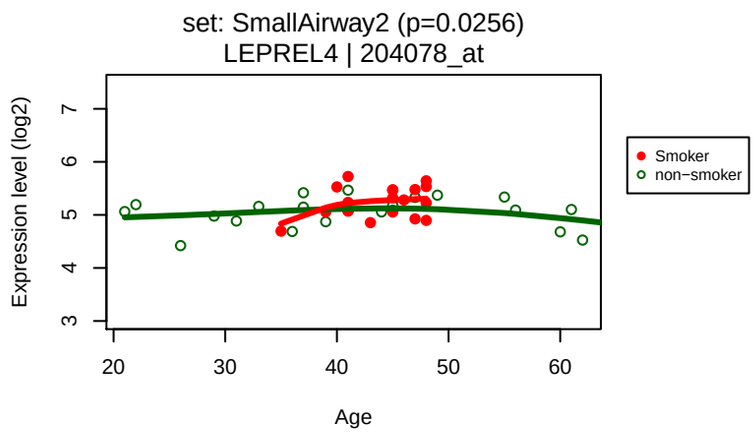
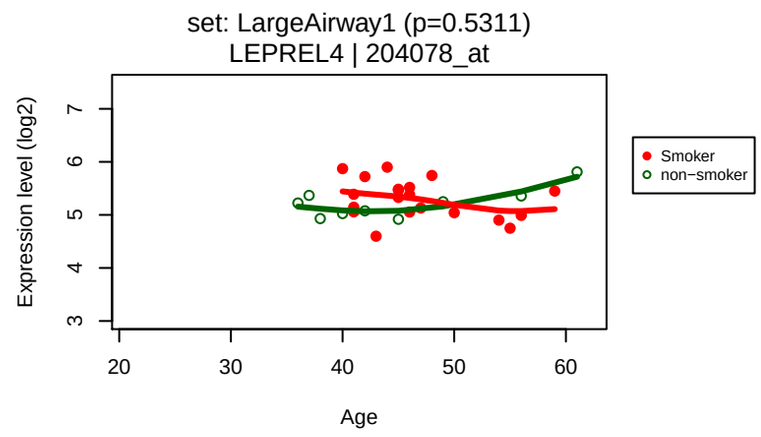

Gene expression of smokers and nonsmokers
gene: LEPREL4 | 204078_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.315
p=0.0256
p=0.584
p=0.5311
p=0.7846
p=0.6673
p=0.1502
p=0.1076
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

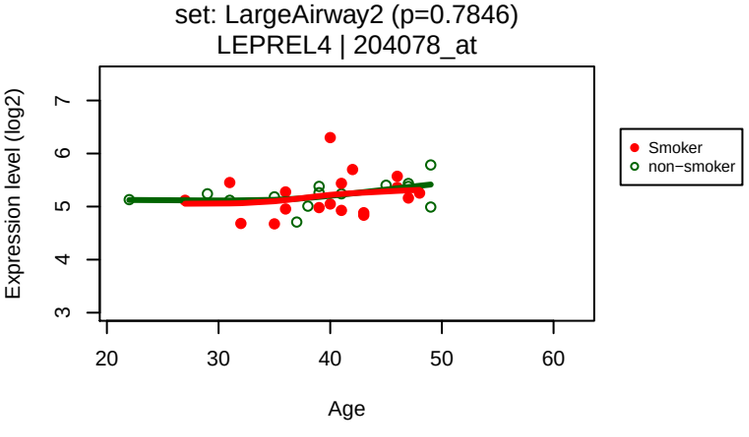

Gene expression ratio (smokers/nonsmokers)
gene: LEPREL4 | 204078_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.315
p=0.0256
p=0.584
p=0.5311
p=0.7846
p=0.6673
p=0.1502
p=0.1076
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

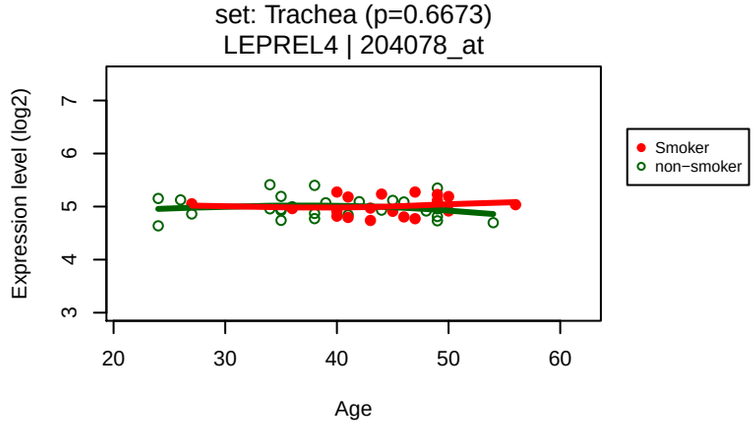

Gene expression ratio in Males
gene: LEPREL4 | 204078_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.011
p=0.8591
p=0.8988
p=0.2124
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LEPREL4 | 204078_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6912
p=0.5163
p=0.9081
p=0.2959
SmallAirway2
LargeAirway2
Trachea
Macrophage2

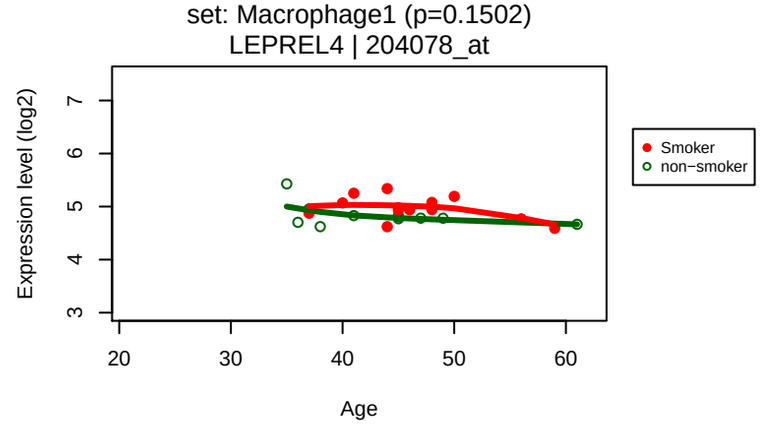

set: SmallAirway1 (p=0.315)
LEPREL4 | 204078_at
Expression level (log2)
Age
Smoker
non−smoker

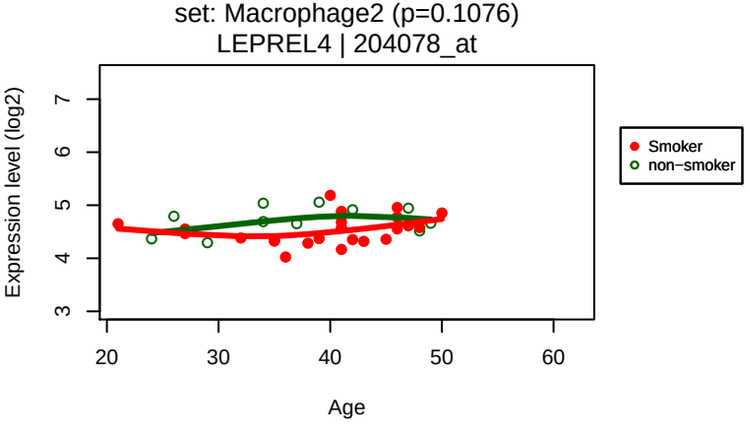

set: SmallAirway2 (p=0.0256)
LEPREL4 | 204078_at
Expression level (log2)
Age
Smoker
non−smoker

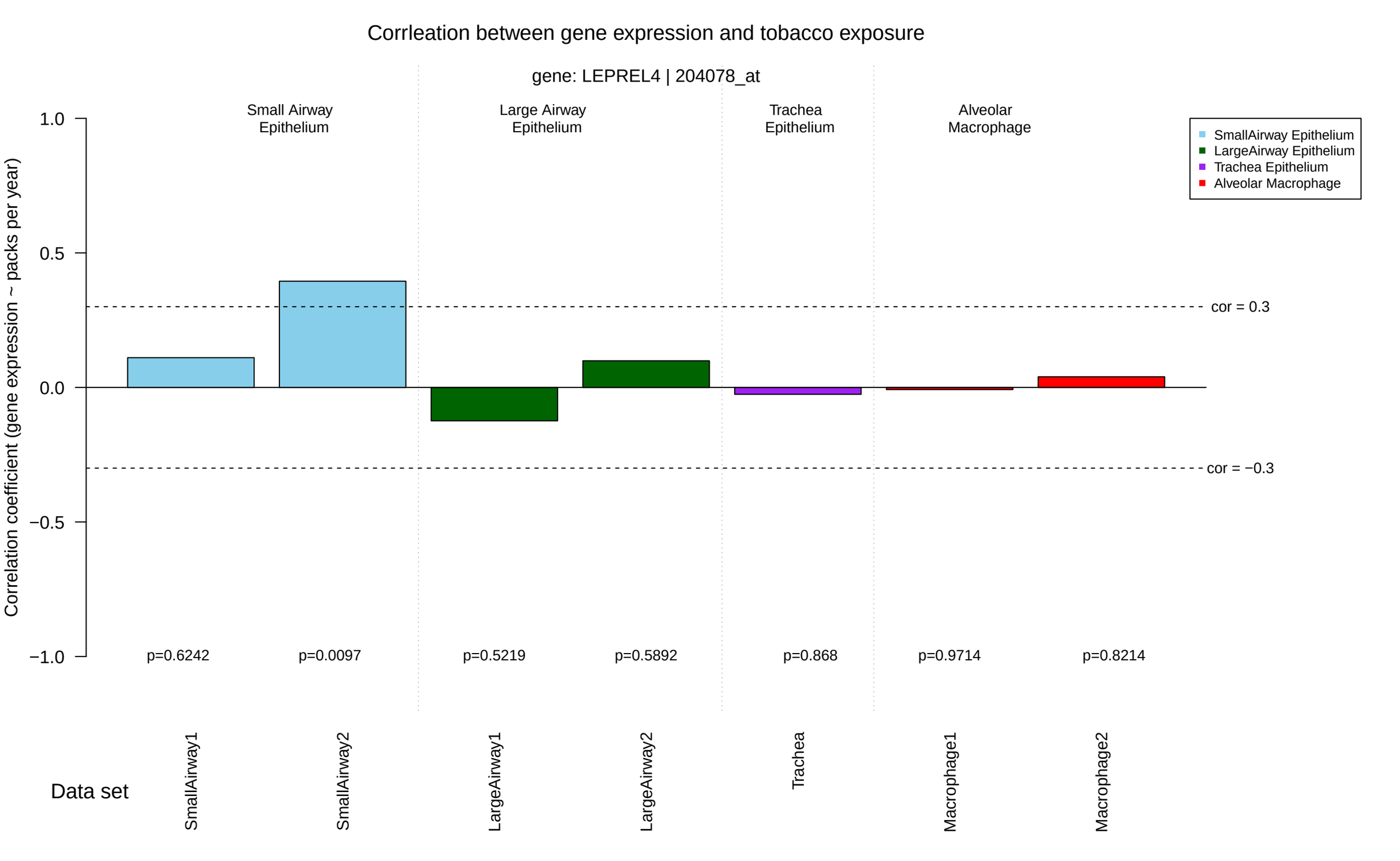

set: LargeAirway1 (p=0.5311)
LEPREL4 | 204078_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7846)
LEPREL4 | 204078_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6673)
LEPREL4 | 204078_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1502)
LEPREL4 | 204078_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1076)
LEPREL4 | 204078_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LEPREL4 | 204078_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6242
p=0.0097
p=0.5219
p=0.5892
p=0.868
p=0.9714
p=0.8214
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2